2025 Tales of Tribute AI Competition

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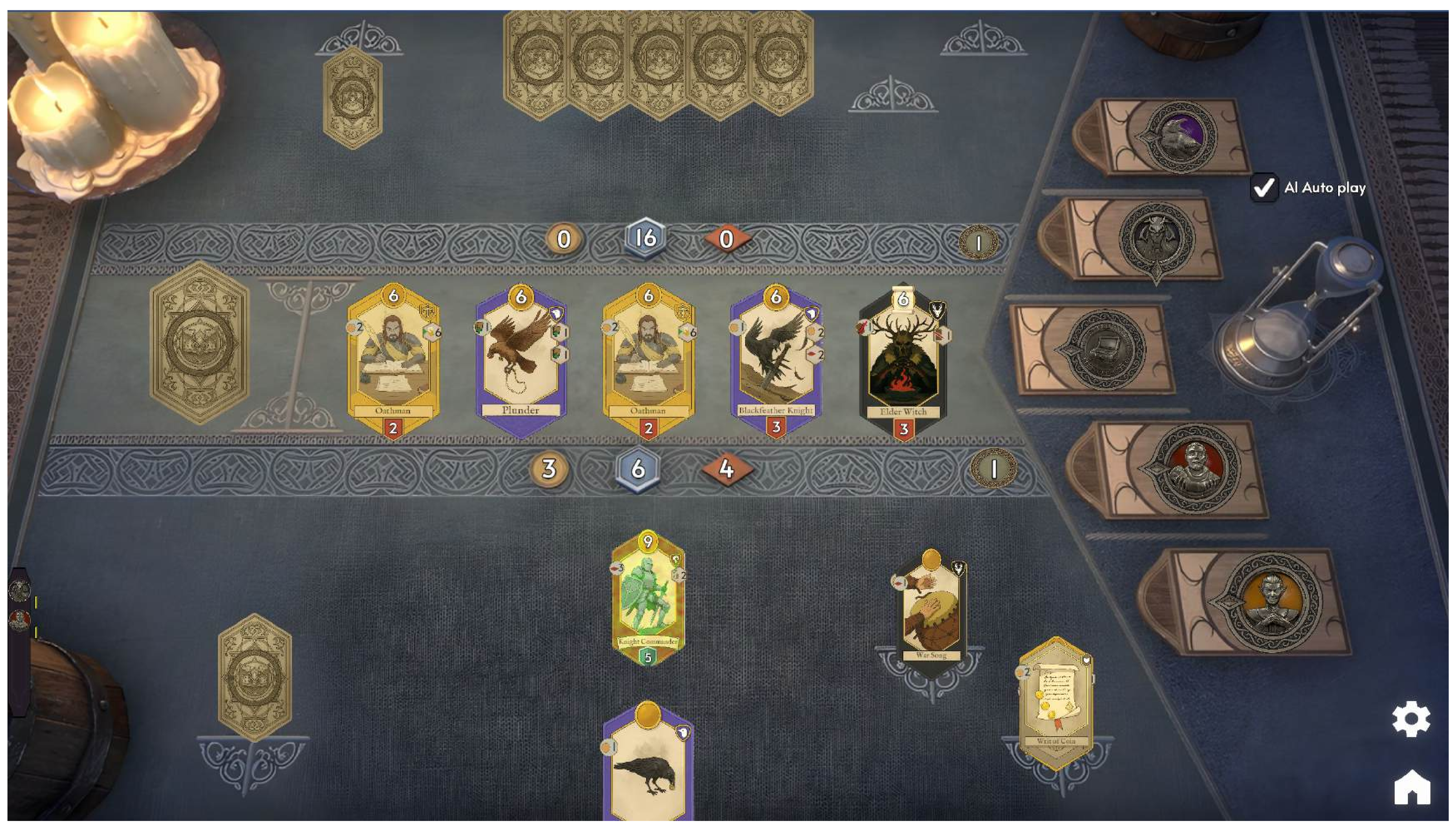


The Challenge

- Large branching factor, multi-action turns
- Advanced deckbuilding
- Lot of hidden information
- Huge amount of nondeterminism
- Yet the gameplay is still very strategical

Competitions

- COG 2023 – 3 entries (+3 baselines)
- COG 2024 – 5 entries
- COG 2025 – 11 entries

The Game

- Implementation of Tales of Tribute card game from The Elder Scrolls® Online
- Adjusted specifically for AI competition
- `gRPC`-based interface, supporting any programming language
- Easy to use Python and C# libraries
- Rich environment, many tools available for developers.
- More than one victory condition
- Deckbuilding done throughout the game
- Wide range of algorithms viable, from MCTS to Neural Networks
- Highly detailed GUI application to play against AI agents
- Cards from the same deck combo together into stronger effects

Cards

The game consists of 8 decks, each deck contains several cards of multiple types:

Actions
Agents
Contract Actions
Contract Agents

There are many effect for each card, like:

Gain resources
Destroy card from your deck
Acquire card

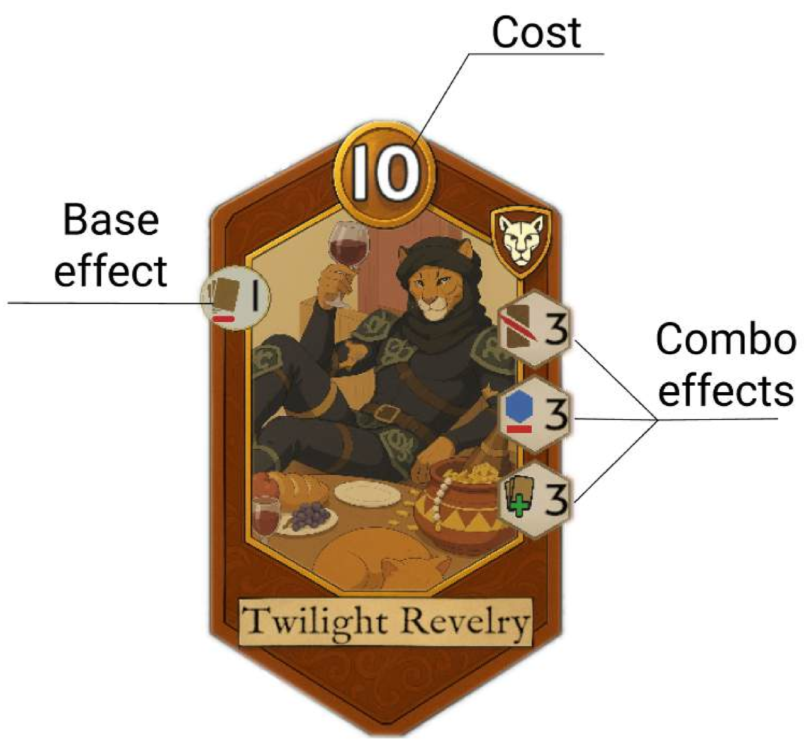
Actions

Draft phase:

- `PICK patron`

Battle phase:

- `PLAY_CARD cardId`
- `ATTACK agentId`
- `ACTIVATE agentId`
- `BUY cardId`
- `MAKE_CHOICE cardIds`
- `END_TURN`



Agents

- **BestMCTS3**, *Adam Ciężkowski, Artur Krzyżyński* (C#)

  MCTS with heuristic function.
  Heuristic function with weights determined by evolutionary algorithm.
  Winner of 2023 & 2024 competition.

- **Council Of Two**, *Thomas van Rossum* (Python 3)

  MCTS and greedy breadth first search.
  Heuristic enhanced.

- **EBot_C**, *David Castejón* (C#)

  Evolutionary techniques.
  Heuristic evaluation function, parameterized by a vector of weights.
  "Coevolution Mode"

- **EBot_F**, *David Castejón* (C#)

  Fixed mode of evolution.

- **HQL_Plus_BOT**, *Helik ThackeR* (C#)

  Double DQN with BeamSearch.
  Based on the agent that achieved second place in 2024.

- **MaltheMCTS**, *Malthe Sørensen* (C#)

  MCTS bot, inspired by AlphaZero.
  Use of small Ensembled Tree Models.

- **MCTS_MMHVR**, *Alessio Iacullo* (Python 3)

  MCTS agent with MinMax hand value rating.

- **NAgent**, *Niklas Angert* (Python 3)

  Neural network–based agent.
  End-to-end policy trained using self-play and Optimistic Smooth Fictitious Play.

- **Sakkirina**, *Katsuki Ohto* (C#)

  MCTS using rule-based policy and value functions.
  Variant of Progressive Widening for stochastic state transition.
  Search the opponent's turn too.

- **SakkirinaSolo**, *Katsuki Ohto* (C#)

  MCTS using rule-based policy and value functions.
  Variant of Progressive Widening for stochastic state transition.
  Search in its own turn.

- **ZMyBot**, *Thomas van Rossum* (Python 3)

  Breadth-first search for prestige.

Future of the competition

- Addition of new decks
- More QOL features
- IEEE COG 2026 (with deck rotation)

Acknowledgments

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